MUKUL PANDEY

Full-Stack Software Engineer | Applied AI & LLM Integration

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Profile

Full-Stack Software Engineer at Volkswagen Group Digital Solutions India with hands-on experience building enterprise applications using **Java, Quarkus, REST APIs, Hibernate/JPA, SQL**, and **Vue.js**. Experienced in backend service development, database integration, API security, automated testing, and microservices maintenance. Builds and deploys full-stack applications using **React, TypeScript, Spring Boot,** and **PostgreSQL**, including AI-powered workflows integrating the **Gemini API** for intelligent issue-draft generation. Combines practical software engineering experience with an academic foundation in machine learning and deep learning.

Skills

Programming Languages: — Java, Python, JavaScript, TypeScript, SQL, C/C++

Backend & Web Development: — Spring Boot, Quarkus, Node.js, Hibernate ORM/JPA, Panache, REST APIs, API Design, React.js, Vue.js, HTML5, CSS3

AI & Machine Learning: — Gemini API, LLM/API Integration, Prompt Engineering, Structured AI Responses, AI-Assisted Workflow Automation, CNN, EfficientNetB0, Logistic Regression, Support Vector Machine (SVM), Hyperparameter Tuning

Databases & Reporting: — PostgreSQL, MySQL, Oracle, MongoDB, Power BI, Power BI Report Embedding, SQL Querying, Data Visualization

Testing & Automation: — Playwright, JavaScript/TypeScript Test Automation, End-to-End Testing, UI Automation, API Testing, JUnit 5, Mockito, Unit Testing, Integration Testing using Docker

Cloud & DevOps: — Docker, OpenShift, Argo CD, Linux, Bash/Shell Scripting, Git, GitHub, Vercel, Render, Neon

Security & Authentication: — JWT, OpenID Connect (OIDC), Role-Based Access Control (RBAC), Row Level Security (RLS), CORS

Software Engineering: — Microservices Architecture, Object-Relational Mapping (ORM), Database Integration, API Integration, Dependency Management, Debugging and Troubleshooting

Experience

Software Engineer, *Volkswagen Group of Digital Solutions India* Oct 2024 – Present | Pune, India

- Develop and maintain Java-based enterprise microservices for the FAP Data Warehouse ecosystem using Quarkus, Hibernate ORM/Panache, REST APIs, and SQL, integrating backend services with Vue.js frontend applications across Jira, KPM, KPMS Halle, and KPMS Feld/NP domains.
- Develop and enhance data retrieval and reporting capabilities, including random-access lazy loading for large datasets and an export service supporting configurable column selection, asynchronous report processing, and downloadable reports.
- Develop an image service that compresses images, stores them in Amazon S3, and provides links for viewing and downloading.
- Implement and maintain scheduled jobs for Jira data ingestion and periodic cleanup of orphaned tickets in the data warehouse.
- Implement authentication and authorization using OpenID Connect (OIDC), role-based access control, and API Helper components to secure service endpoints and support authenticated access.
- Manage environment-specific configurations and deployment workflows using Helm charts, Argo CD, and OpenShift; contribute to dependency upgrades, troubleshooting, and service reliability.
- Develop and maintain unit and integration tests using JUnit 5 and Mockito, and contribute to code quality through SonarQube analysis, test coverage reporting, and CI/CD quality checks.
- Recognized through VWGDS awards for contributions to the API Helper and Export Service initiatives, including the TrendSetter, Innovation, and InstaPat awards.

Projects

FAP Data Ware House, *Currently working as full stack java developer,* Mar 2025 – Present
Java, Quarkus, Vue, Jira, Docker, Hibernate, Sql, Orm, Jwt, Mockito, Junit 5

DevTrack — AI-Powered Project and Issue Management Application Sep 2026
Technologies: React, TypeScript, Java 21, Spring Boot, PostgreSQL, Gemini API, REST APIs, Vercel, Render, Neon

- Designed, developed, and deployed a full-stack application for managing projects, issues, members, labels, assignees, comments, and issue-status history.

- Developed REST APIs using Spring Boot for authentication, project management, issue tracking, and related workflows, backed by PostgreSQL.
- Built a responsive frontend using React and TypeScript and integrated it with the backend APIs.
- Integrated the Gemini API to power an AI Issue Assistant that transforms rough issue descriptions into structured drafts containing a title, detailed description, acceptance criteria, and suggested labels.
- Designed the AI-assisted workflow to let users review and edit generated content before explicitly creating an issue, keeping AI generation separate from database persistence.
- Deployed the frontend on Vercel, backend on Render, and database on Neon.

Deep Learning based Disease Detection System for Plants Jan 2024 – Jun 2024

Technologies: Python, CNN, EfficientNetB0, Streamlit, hyperparameter tuning

Developed a deep-learning-based plant disease detection application using CNN-based image classification and EfficientNetB0.

- Applied parameter tuning to improve model performance and built an interactive application using Streamlit.
- Reported 99.86% accuracy in the project evaluation; clarify the evaluation dataset and methodology when presenting this result.

Cancer cell type prediction & House price prediction 2022 – 2024

Technologies: Python, Logistic Regression, Support Vector Machine (SVM)

Built machine-learning classification models using Logistic Regression and SVM for cancer cell-type prediction.

- Explored model performance and parameter tuning to evaluate classification results.

Antardarshnam (Smart India Hackathon 2022) Sep 2021 – Jan 2022

Technologies: React, JavaScript, HTML, CSS, Django, MongoDB

- Contributed to a web platform offering educational and self-awareness resources, including books, games, and videos.
- Grand Finalist, Smart India Hackathon 2022.

Education

Master of Business Administration Jan 2025 – Present | Pune, India
Indira Gandhi National Open University, Regional Centre Pune

PG-Diploma in Big Data Analytics, *CDAC Pune* Aug 2024 – Sep 2024 | Pune, India

B.Tech CSE, *Institute of Engineering and Rural Technology Prayagraj* Nov 2020 – Jul 2024 | Prayagraj, India
I-DIV with distinction. CGPA=8.19

10+2 with PCM and Computer Science, *KV Kanpur Cantt.* Apr 2018 – May 2019 | Kanpur, India

- 10 CGPA in Class 10th
- 96.2 % in Class 12th

Certificates

- AWS certified in Blu Age L2 Certification
- Google ML Bootcamp India
- Google Cloud Certified Professional Data Engineer
- NASSCOM certified in Fundamentals of Digital Marketing using Social Media Platforms

Awards

TrendSetter Award , *VWGDS*
For the development of API Helper

InstaPat Award , *VWGDS*
For the development of Export Service

Innovation Award , *VWGDS*
For the development of API Helper

GATE 2024: (DA) rank 5,322; CS/IT rank 10,246. , *NTA*

Grand Finalist, Smart India Hackathon 2022. , *MIC*

Top 10 Finalist, HackCBS 5.0. , *SSCBS*

Winner, NOVATHON 2022. , *BBDEC*